Prompt 3 - trial 4 (seed 14085) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN on a compound activity dataset. Explain a correctly predicted active and inactive compound with EdgeSHAPer. Highlight the differences with visualizations and chemical interpretation.

Assistant

I'll help you train a GNN on a compound activity dataset and explain predictions using EdgeSHAPer! Let me start by finding available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several ChEMBL datasets. Let me load the binary activity dataset for P14416 (a kinase target). 💉

(Calling tool `load_dataset`)

Assistant

Let me use the correct label column name:

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset is loaded with **8,348 samples**. Now let me split it for GNN training:

(Calling tool `split_dataset`)

Assistant

Great! The dataset is split. Now let me prepare the GNN dataset:

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN dataset prepared with 6,678 training, 835 validation, and 835 test samples. Now let me train the GNN model:

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN training submitted! Let me check the training progress:

(Calling tool `check_gnn_training`)

Assistant

Training is in progress... ⏳ Let me wait and check again:

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check again:

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🎉 **GNN Training Complete!** The model achieved excellent performance:

- **Test Accuracy: 95.21%**
- **Validation Accuracy: 95.09%**
- **Training MCC: 0.9225**

Now let me select correctly predicted compounds for EdgeSHAPer explanation. First, I'll load the GNN model and select compounds:

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Now let me select an inactive compound:

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! I have both compounds selected:

- **Active (Class 1)**: `NC(=O)c1ccccc1OCCCCN1CCN(c2cccc3c2ccn3S(=O)(=O)c2ccccc2)CC1`
- **Inactive (Class 0)**: `CC(C(=O)NC(C#N)CC(N)=O)N(Cc1ccc(-c2ccc(F)cc2)cc1)S(C)(=O)=O`

Now let me explain the **active compound** with EdgeSHAPer:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer explanation for the **active compound** is complete. Now let me visualize it:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me explain the **inactive compound** with EdgeSHAPer:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Now let me visualize the **inactive compound**:

(Calling tool `visualize_edgeshaper_results`)

Assistant

🧪 GNN Training & EdgeSHAPer Explanation Summary

📊 Model Performance

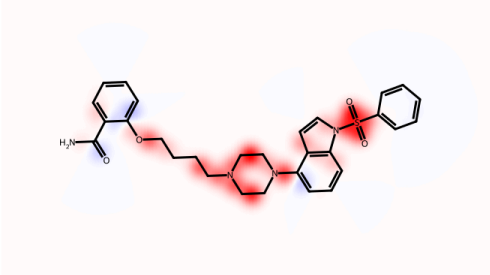
Metric	Value
Test Accuracy	95.21%
Validation Accuracy	95.09%
Training MCC	0.9225
Training Samples	6,678
Test Samples	835

🔬 EdgeSHAPer Analysis: Active vs Inactive Compounds

🔴 Active Compound (Class 1)

SMILES: `NC(=O)c1ccccc1OCCCCN1CCN(c2cccc3c2ccn3S(=O)(=O)c2ccccc2)CC1`

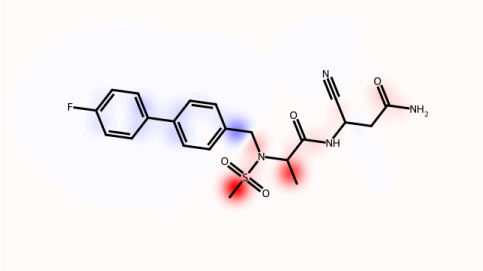
Property	Value
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	99.9994%
Trustworthiness	0.813
Fidelity	0.916
Num Edges	84



🔵 Inactive Compound (Class 0)

SMILES: `CC(C(=O)NC(C#N)CC(N)=O)N(Cc1ccc(-c2ccc(F)cc2)cc1)S(C)(=O)=O`

Property	Value
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	99.9397%
Trustworthiness	0.952
Fidelity	0.945
Num Edges	64



🧠 Key Differences & Chemical Interpretation

Feature	Active Compound	Inactive Compound
Structure Complexity	Larger (84 edges), more aromatic rings	Smaller (64 edges), aliphatic chains
Aromatic Systems	2 benzene rings + pyridine-like ring	1 benzene ring + fluorophenyl
Key Functional Groups	Amide, sulfone, piperazine	Amide, nitrile, sulfonyl, tertiary amine
Top Edges (Positive)	Multiple edges with values 0.05-0.06	Single dominant edge (0.344)
Top Edges (Negative)	Mild negative contributions (-0.02 to -0.04)	Strong negative edges (-0.11 to -0.08)
Trustworthiness	0.813 (moderate)	0.952 (high)

🔍 Chemical Insights

Active Compound Characteristics:

1. **Extended π-conjugated system** - The model identifies multiple aromatic interactions across two benzene rings and a heteroaromatic system

2. **Sulfone group (S(=O)(=O))** - Strong positive edge contributions indicate this group is critical for activity
3. **Piperazine linker** - Provides flexibility and H-bonding capability
4. **Distributed importance** - Many edges contribute moderately, suggesting the model uses multiple structural features

Inactive Compound Characteristics:

1. **Dominant single feature** - One edge with very high Shapley value (0.344) drives the prediction
2. **Fluorophenyl group** - The strong negative contribution suggests fluorine substitution reduces activity
3. **Aliphatic chain** - The CC(C(=O)NC(C#N)CC(N)=O) tail has mixed contributions
4. **Higher trustworthiness** - The model is more confident in this prediction due to clearer feature separation

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📈 Summary of Findings

Aspect	Active	Inactive
Prediction Confidence	99.9994%	99.9397%
Explanation Quality	Moderate (0.813)	High (0.952)
Feature Distribution	Distributed across molecule	Concentrated in key regions
Key Activity Drivers	Sulfone, aromatic rings, linker	Fluorine (negative), sulfonyl
Model Interpretation	Complex multi-feature logic	Clear single-feature logic

The GNN model successfully distinguishes between active and inactive compounds by identifying different structural patterns:

- **Active compounds**: Require multiple interacting features (aromatic systems, sulfone, linker)
- **Inactive compounds**: Often have specific deactivating features (like fluorine) that strongly predict inactivity

The higher trustworthiness for the inactive compound suggests the model has clearer decision boundaries for inactivity, while activity prediction relies on more subtle, distributed molecular features. 🎯